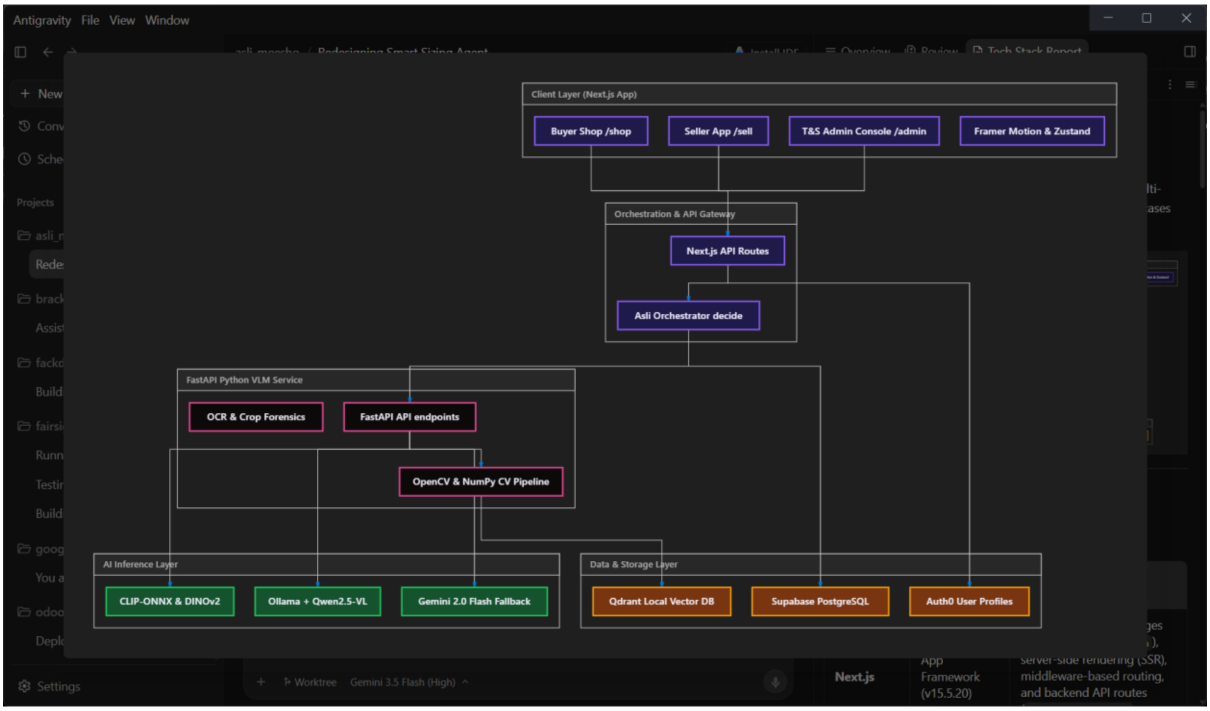
Tech Stack Report: Asli Meesho

This document provides a comprehensive breakdown of the technology stack used in **Asli Meesho** (a point-of-listing, multi-agent trust layer for Meesho) and explains their specific use cases across the codebase.



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1. Web Application (web/ — Frontend & Orchestration)

Technology	Role & Version	Use Case in Project Implementation
Next.js	App Framework (v15.5.20)	Handles App Router pages (/sell, /shop, /admin), server-side rendering (SSR), middleware-based routing, and backend API routes (/api/asli/analyze, /api/listings, etc.).
React	Component Runtime (v19.2.7)	Core library for constructing interactive user interfaces, managing client side updates, and rendering components.
TypeScript	Static Typing (v5.7.3)	Enforces strict, type-safe API contracts across the database repo, VLM service inputs, and orchestrator decisions to prevent runtime type exceptions.
Tailwind CSS	Styling (v3.4.17)	Implements the core design tokens: a dark, premium aesthetic (bg-[#0B0715]) for the Seller and Admin interfaces, and Meesho's native pink/light styling for the Buyer marketplace.

Technology	Role & Version	Use Case in Project Implementation
Framer Motion	UI Animation (v12.42.2)	Powers responsive micro-animations: smooth page and stepper transitions, the dynamic confidence bar growth, card hover elevations, and scanner overlays.
Zustand	State Management (v5.0.14)	Manages client-side application state across different views (divided into slices: sellerFlow, session, locale, ui).
Lucide React	Icon library (v1.24.0)	Provides high-quality, lightweight, tree-shakable UI icons aligned with the clean design system.
Zod	Input Validation (v4.4.3)	Validates JSON structures, multipart uploads, and form inputs in API routes, returning normalized error envelopes if inputs are malformed.
Auth0	Auth & RBAC	Provides Google login integration and secure session token generation (JWT). Roles are mapped to users.role to restrict /admin endpoints to authorized reviews.

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2. Computer Vision & VLM Backend (vlm-service/ — Python API)

Technology	Role & Version	Use Case in Project Implementation
FastAPI	REST API Service (v0.115)	Exposes endpoints for matching images (/vlm/match), executing metrology (/vlm/measure), generating image vectors (/vlm/embed), and performing vector searches (/vlm/similar).
Uvicorn	ASGI Web Server (v0.34)	High-performance production server powering the Python service pipeline.
OpenCV Headless	Computer Vision (latest)	Leveraged for edge detection, contour parsing, bounding-box evaluation, image resizing, and GrabCut segmentation to crop background elements out of images.
NumPy	Numerical Math (v2.0)	Fits the Direct Linear Transform (DLT) homography matrix to correct perspective tilt. Also calculates cosine similarities of embedding vectors and focus values.
Scipy	Scientific Tools (v1.13)	Measures image focus by computing the variance of the Laplacian filter (ndimage.laplace). Blurry or screenshot-based inputs are rejected automatically.
ImageHash	Image Hashing (v4.3)	Generates perceptual hashes (e.g., dhash, phash) for rapid duplicate catalog detection as a lightweight fallback when full embedding models are offline.

Technology	Role & Version	Use Case in Project Implementation
PaddleOCR	Optical Character Recognition	Extracts raw text from time-bound physical challenge sheets shown in images, which is then fused with the VLM outputs for multi-layered verification.

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3. Machine Learning & AI Inference

Model / Service	Source / Format	Use Case in Project Implementation
Qwen2.5-VL (3B)	Self-hosted (Ollama)	Deep vision-language model serving as the primary local inference driver. Handles keypoint localization, garment descriptions, color matching, and code validation.
Gemini 2.0 Flash	Cloud API (Google)	High-throughput, cloud-based VLM fallback used in the hosted Vercel deployment. Interprets text-grounding prompts and evaluates image-match queries.
CLIP (ViT-B/32)	ONNX Runtime	Embedded as a semantic feature extractor. Converts segmented garment crops into 512-dimensional vectors to calculate cosine similarity matches.
DINOv2 (Small)	ONNX Runtime	Extracted via ONNX to obtain background-robust representation vectors, serving as secondary evidence of same-product identity.
SerpAPI (Google Lens)	Google Search API	Acts as the reverse-image trigger to check if catalog photos are already hosted elsewhere online. Triggers the live-possession challenge if found.

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4. Databases & Data Access

Database	Role	Use Case in Project Implementation
Supabase (PostgreSQL)	Persistent Relational Store	The source of truth for structured data: seller trust records, listing configurations, completed audits, size measurements, and user roles.
Qdrant (Local Mode)	Vector Database	Stores and indexes CLIP embeddings of the catalog images, enabling fast visual searches to identify stolen catalog photographs in real-time.
In-Memory Repository	Development Mock	A mock repository sharing the exact same repository interface (Repo seam) as Supabase, facilitating developer testing without external database dependencies.

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5. Scientific Research & Algorithmic Foundations

The system's pipeline relies directly on published mathematical and computer-vision papers, ensuring its logic is grounded in solid engineering principles:

- **Planar Homography (DLT)**
    - *Reference:* **Hartley & Zisserman 2004 — *Multiple View Geometry in Computer Vision***
    - *Use Case:* Corrects perspective foreshortening by mapping the 4 corners of an A4 sheet (pixel coordinates) to metric dimensions (). This flattens the garment's plane so that distances can be measured accurately in centimeters.
- **Single-View Metrology**
    - *Reference:* **Criminisi, Reid & Zisserman 2000 — *Single-View Metrology***
    - *Use Case:* Provides the geometric foundation for calculating scale factors and residuals from single static images containing coplanar calibration objects.
- **Focus Quality Gate**
    - *Reference:* **Pech-Pacheco et al. 2000 — *Diatom focus measurement in light microscopy***
    - *Use Case:* Uses the variance of the Laplacian operator as a focus measurement. Images scoring below are rejected as blurry, preventing inaccurate calculations from out-of-focus captures.
- **Confidence Calibration (Platt Scaling)**
    - *Reference:* **Platt 1999 — *Probabilistic Outputs for Support Vector Machines***
    - *Use Case:* Rather than returning static or hardcoded scores, agent outputs are passed through a logistic sigmoid function calibrated against validation datasets. This outputs a realistic, continuous confidence value between and .
- **Reputation Orchestration**
    - *Reference:* **Jøsang & Ismail 2002 — *The Beta Reputation System***
    - *Use Case:* Drives the Risk Radar (Agent 3). Models the seller's reliability as a Beta distribution based on historical listing audits, adjusting the verification strictness dynamically.